

ECE 4644 / 5610 Satellite Communications**Homework Assignment #5 - DEMO**

Note: For all problems take the radius of Earth, R_E , to be 6378.137 km.

Problem #1

A satellite receiver system operates on 30 GHz. The satellite receiving antenna has a noise temperature of 75 K and the LNA has a noise temperature of 100 K with a gain of 40 dB. The mixer section has a noise temperature of 500 K and a gain of 5 dB; the IF section has a noise temperature of 1000 K and a gain of 20 dB.

- a. Determine the system noise temperature.
- b. Someone suggests improving the noise situation in the receiver by reducing the noise temperature of one of the stages. The same cost applies to reducing the noise temperature of the LNA by 25 K (RF stage), of the mixer by 100 K, and of the IF stage by 500 K. Make a recommendation and briefly explain your reasoning. The listener is skeptical - acknowledge the reasons for his/her skepticism, then proceed with your explanation.

Problem #2

A receiver system needs to have its antenna mounted out-of-doors. A length of cable with a loss of 3 dB and physical temperature of 300° K is used to connect the antenna to the high-gain LNA and the rest of the receiver. The antenna and the LNA have the same noise temperature, 25 K. Determine the system noise temperature and the increase in receiver noise power in dB due to insertion of the cable. (Remember that the system noise temperature is referenced to the output terminals of the antenna, i.e., before the run of cable.)

Problem #3

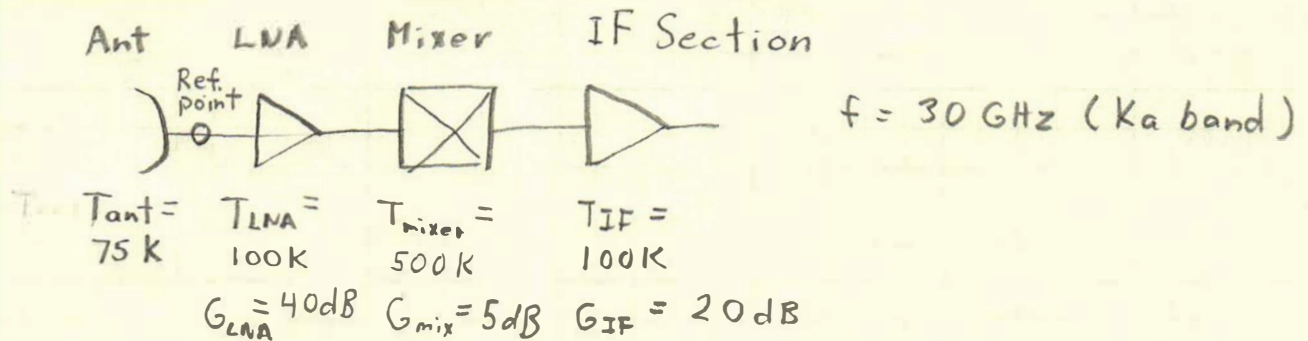
We will design a transmitting earth station to work with a Starlink-type satellite system working in Ku band. The aim is to produce a C/N of 20 dB in the satellite transponder at a frequency of 14.018 GHz. The satellite is in a circular orbit at an altitude of 600 km. The uplink antenna has dimensions of 30 cm by 25 cm and an aperture efficiency of 50%. The receive system noise temperature is 250 K and the noise bandwidth is 36 MHz. The satellite receiving antenna has a gain of 25 dB.

- a) If the uplink is expected to reach threshold $(C/N)_{up}$ when the satellite is at an elevation angle of 45°, estimate (or solve exactly) the maximum range of operation
- b) Determine the noise power at the input to the transponder
- c) Determine the transmitter power that is needed to achieve the desired $(C/N)_{up}$. To this point assume no losses apart from the inevitable path loss

- d) Now we will introduce some realism. Neither antenna will have its axis directed at the other (probably). Allow for 3 dB of loss for edge-of-beam at both ends of the link. Allow 3 dB for propagation loss in the atmosphere and 1 dB of miscellaneous loss. Now determine the transmitter power that is required.

Prob. #1

Satellite Rx System



a) The system noise temperature is found from

$$T_s = T_{\text{ant}} + T_{\text{LNA}} + \frac{T_{\text{mixer}}}{G_{\text{LNA}}} + \frac{T_{\text{IF}}}{G_{\text{LNA}} G_{\text{mixer}}}$$

Convert gains to Linear units:

$$G_{\text{LNA}} = 40 \text{ dB} = 10,000, G_{\text{mixer}} = 5 \text{ dB} = 3.16, G_{\text{IF}} = 20 \text{ dB} = 100$$

$$\text{Then } T_s = 75 + 100 + \frac{500}{10,000} + \frac{100 \text{ K}}{(10,000)(3.16)} = \underline{175.08 \text{ K}}$$

> Observe that with G_{LNA} so high that the mixer and IF section make almost no contribution to T_s .

b) Reducing the noise temperature will produce these results:

$$T_{\text{LNA}} \downarrow 25 \text{ K} \Rightarrow T_s \downarrow 25 \text{ K}$$

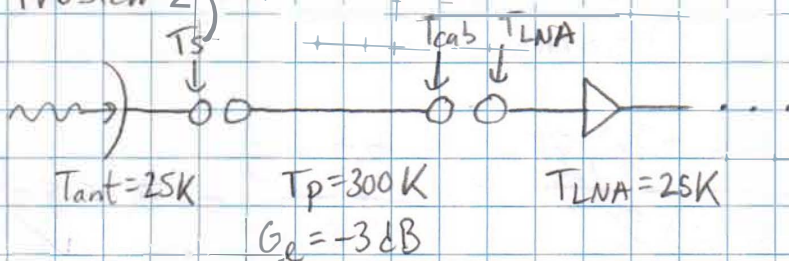
$$T_{\text{mixer}} \downarrow 100 \text{ K} \Rightarrow T_s \downarrow \frac{100}{10,000} = 0.01 \text{ K}$$

$$T_{\text{IF}} \downarrow 500 \text{ K} \Rightarrow T_s \downarrow \frac{500}{(10,000)(3.16)} = 0.016 \text{ K}$$

> It is better to reduce the noise temperature of the LNA by 25 K than to reduce the temperatures of the other components by much larger amounts.

Explanation: While it is tempting to think that large decreases in temperature would be better, it is the noise that appears at the front of the receiver that is boosted by all the gain of the receiver, and it is there - at the antenna and LNA - that we should devote our attention.

Problem 2)



Given that there is a lossy cable between the antenna and LNA, I expect the noise temperature and noise power to increase. As a baseline, let's find the system noise temperature with no cable.

$$T_s \approx T_{ant} + T_{LNA} = 50K$$

We can now find the system temperature with the cable and compare. T_{cab} can be solved for using the lossy medium equation:

$$T_{cab} = T_p(1 - G_e) = 300(1 - 10^{-3/10}) = 149.6K \quad \checkmark$$

The noise temperature at the input to the LNA is:

$$T_{LNA}^{in} = T_{cab} + T_{LNA} + (0.5)T_{ant}$$

$$= 149.6K + 25 + (0.5)25 = 187.1K$$

T_{ant} is multiplied by 0.5 because the noise power contributed by the antenna is dropped by 3 dB after passing through the lossy cable.

Given T_{LNA}^{in} , we can determine T_s at the correct reference point (output of antenna) by reversing the effect of the cable loss.

$$T_s = 2 \times T_{LNA}^{in} = 2(187.1K) = \boxed{374.3K} \quad \checkmark$$

$$T_{S, \text{nocable}} = 50 \text{ K} \quad T_{S, \text{cable}} = 374.3 \text{ K}$$

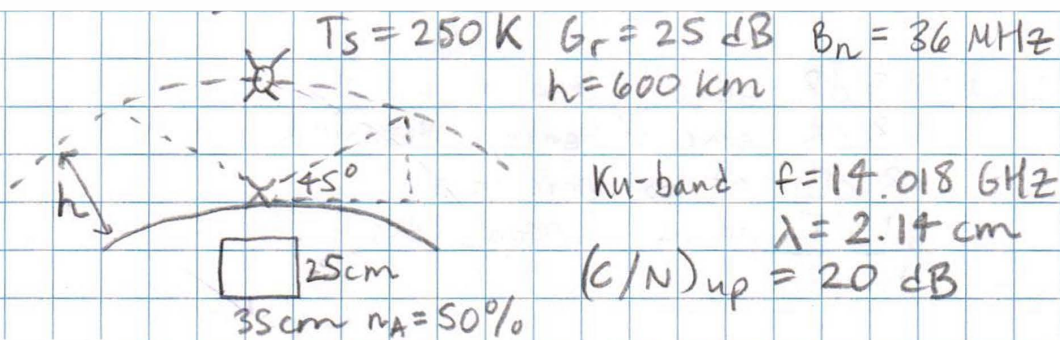
To find the increase in noise power N , we can use the KTB relationship. (Assume B does not change)

$$\frac{N_{\text{cable}}}{N_{\text{nocable}}} = \frac{k(374.3 \text{ K})B}{k(50 \text{ K})B} = \frac{374.3}{50} = 7.48 = \boxed{8.74 \text{ dB}}$$

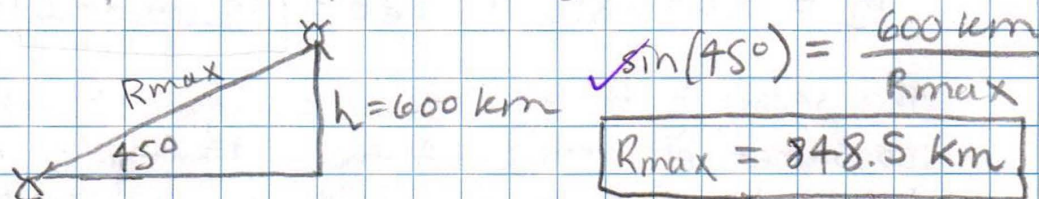
The noise power increased by a factor of 7.48 (8.74 dB) by inserting a lossy cable. This demonstrates the drastic negative affects of allowing losses before the LNA.

As discussed in class, we want to minimize the run of cable between the antenna and LNA by attaching a low noise block converter at the antenna. Without doing this, the noise power may increase enough to cause an outage or prevent the system from functioning in the first place.

Prob. #3



- a) Assuming a locally flat Earth, we can find the maximum range of operation by considering the right triangle:



- b) Using $N = KTB$ at the input to the transponder:

$$N = (1.38 \times 10^{-23}) (250) (36 \times 10^6) = 1.242 \times 10^{-13} \text{ W}$$

$$N = -129.06 \text{ dBW}$$

- c) For $(C/N)_{up} = 20 \text{ dB}$, find $P_t = ?$ $L_{tot} = 0 \text{ dB}$

$$G_t = n_A \frac{4\pi A_p}{\lambda^2} = (0.50) \cdot \frac{4\pi (0.25 \text{ m})(0.35 \text{ m})}{(0.0214 \text{ m})^2} = 1200.5$$

$$= 30.79 \text{ dB}$$

$$L_p = \left(\frac{4\pi R_{max}}{\lambda} \right)^2 = \left(\frac{4\pi \cdot 848.5 \cdot 10^3 \text{ m}}{0.0214 \text{ m}} \right)^2 = 2.48 \times 10^{17}$$

$$= 173.94 \text{ dB}$$

Link	
P_t	$P_t \text{ dBW}$
G_t	30.79 dB
G_r	25 dB
L_p	-173.94 dB
P_r	$P_t - 118.15$

$$\rightarrow -129.06 \text{ dBW}$$

$$(C/N)_{up} = 20 \text{ dB} \Rightarrow P_r = N + 20 \text{ dB} = P_t - 118.15 \text{ dB}$$

$$\Rightarrow P_t = N + 20 \text{ dB} + 118.15 \text{ dB} = 9.09 \text{ dBW} = 8.12 \text{ W}$$

d) Assuming additional losses to promote realism:

$$\begin{array}{rcl} L_{\text{tot}}: & 3 \text{ dB} & \text{edge of beam, uplink} \\ & 3 \text{ dB} & \text{edge of beam, downlink} \\ & 3 \text{ dB} & \text{atmospheric loss} \\ & 1 \text{ dB} & \text{miscellaneous loss} \\ \hline & 10 \text{ dB} & \end{array}$$

Considering these losses, the transmit power would need to increase by 10 dB to compensate.

$$P_t' = 9.09 \text{ dBW} + 10 \text{ dB} = \boxed{19.09 \text{ dBW}} = 81.1 \text{ W}$$

This seems to be a reasonable figure to ask of a consumer electronic. A typical lightbulb requires around 100 W and our transmit power is less than this. A quick online search reveals that the average cost per watt is around 0.12 dollars/kWh. Assuming this figure:

$$81.1 \text{ W} = 0.0811 \text{ kW}$$

$$1 \text{ day} = 24 \text{ hrs}$$

$$(0.0811 \text{ kW})(24 \text{ hrs}) = 1.9464 \text{ kWh}$$

$$(1.9464 \text{ kWh})(0.12 \text{ dollars/kWh}) = 0.2335 \text{ dollars}$$

This means that it costs around 23 cents (about a quarter a day) to generate the power required for the consumer transmitter. This is a very plausible figure that could entice customers to utilize the service.